Applied Linear Algebra in Data Analysis

Eigenvalues and Eigenvectors

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Part I

Representation of Linear Transformations are Basis Dependent

Representation of Linear Transformations require a Basis

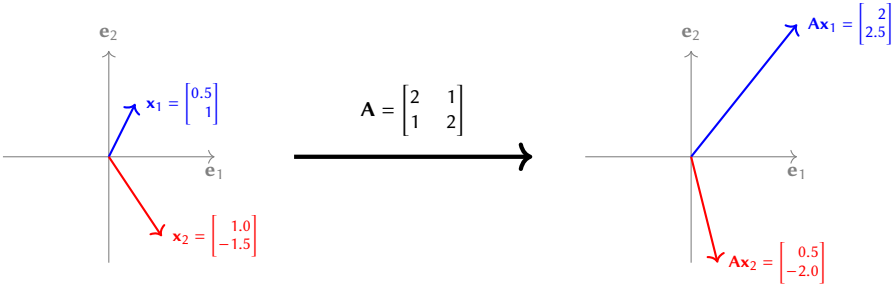
Linear Transformations

Matrices represent linear transformations, $\mathbf{A} \in \mathbb{R}^{p \times q}$ represents a transformation $T : \mathbb{R}^q \longrightarrow \mathbb{R}^p$.

$$\mathbf{y} = T(\mathbf{x}) = \mathbf{A}\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^q \text{ and } \mathbf{y} \in \mathbb{R}^p$$

Consider a linear transformation $T : \mathbb{R}^n \longrightarrow \mathbb{R}^n$.

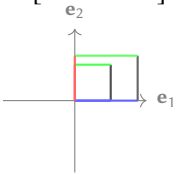
In general, T scales, and rotates/reflects the vector $\mathbf{x}$ to produce $\mathbf{y}$.



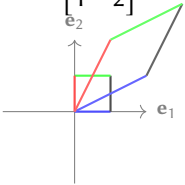
Linear Transformations

An easier way is to look at what happens to the standard basis $\{\mathbf{e}_i\}_{i=1}^{n}$.

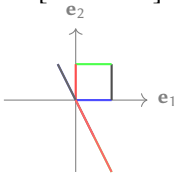
$$\mathbf{A} = \begin{bmatrix} 1.75 & 0 \\ 0 & 1.25 \end{bmatrix}$$



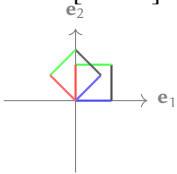
$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



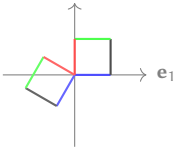
$$\mathbf{A} = \begin{bmatrix} -0.5 & 1 \\ 1 & -2 \end{bmatrix}$$



$$\mathbf{A} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$



$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Linear Transformation in Different Basis

Consider a basis $\mathcal{V} = \{\mathbf{v}_i\}_{i=1}^n$ for $\mathbb{R}^n$. Let $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^\top \in \mathbb{R}^n$ be the representation of $\mathbf{x}$ in the standard basis.

Representation of $\mathbf{x}$ in $\mathcal{V}$ is,

$$\mathbf{x} = \sum_{i=1}^{n} x_{vi}\mathbf{v}_i, \quad \mathbf{x}_{\mathcal{V}} = \begin{bmatrix} x_{v1} & x_{v2} & \cdots & x_{vn} \end{bmatrix}^\top$$

We can go back and forth between these two representations in the following way,

$$\mathbf{x} = \mathbf{V}\mathbf{x}_{\mathcal{V}} \quad \text{and} \quad \mathbf{x}_{\mathcal{V}} = \mathbf{V}^{-1}\mathbf{x}; \quad \text{where, } \mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$$

When $\mathcal{V}$ is an orthonormal basis, then the algebra gets simpler,

$$\mathbf{x} = \mathbf{V}\mathbf{x}_{\mathcal{V}} \quad \text{and} \quad \mathbf{x}_{\mathcal{V}} = \mathbf{V}^\top\mathbf{x}$$

Linear Transformation in Different Basis

Consider a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$.

Consider a vector $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. What is $\mathbf{y} = \mathbf{A}\mathbf{x}$?

$$\mathbf{y} = \mathbf{A}\mathbf{x} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

Now, consider a basis $\mathcal{V} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ for $\mathbb{R}^2$. What is the representation of $\mathbf{x}, \mathbf{y}$ in $\mathcal{V}$?

Linear Transformation in Different Basis

Representation of $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{V}$ is given by,

$$\mathbf{x}_{\mathcal{V}} = \mathbf{V}^{-1}\mathbf{x} = \frac{1}{3}\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\mathbf{y}_{\mathcal{V}} = \mathbf{V}^{-1}\mathbf{y} = \frac{1}{3}\begin{bmatrix} 8 \\ -1 \end{bmatrix}$$

Now, if we apply the linear transformation T on $\mathbf{x}_{\mathcal{V}}$ will we get $\mathbf{y}_{\mathcal{V}}$?

$$\mathbf{A}\mathbf{x}_{\mathcal{V}} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\frac{1}{3}\begin{bmatrix} -4 \\ 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -4 \\ 5 \end{bmatrix} \neq \mathbf{y}_{\mathcal{V}}$$

Representation of a linear transformation T is basis dependent!

Similarity transformation

Linear transformation represent geometric operations on vectors. The representation of these geometric operations through matrices is basis dependent.

A linear transformation represented in one basis represent a different transformation in another basis.

If we have the representation of a linear transformation in one basis, we can find its representation in another basis.

We just need to keep track of the basis one is working in.

Let $\mathbf{x}, \mathbf{y}$ be representations in the standard basis. Changing basis to $\mathcal{V}$, gives us $\mathbf{x}_{\mathcal{V}}, \mathbf{y}_{\mathcal{V}}$.

$$\mathbf{y}_{\mathcal{V}} = \mathbf{V}^{-1}\mathbf{y} = \mathbf{V}^{-1}\mathbf{A}\mathbf{x} = \mathbf{V}^{-1}\mathbf{A}\mathbf{V}\mathbf{x}_{\mathcal{V}} = \mathbf{A}_{\mathcal{V}}\mathbf{x}_{\mathcal{V}}$$

Similarity transformation

Two matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ are called **similar** matrices, if there exists a non-singular matrix $\mathbf{Q}$, such that,

$$\mathbf{B} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}$$

The transformation represented by $\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}$ is called the **similarity transformation**.

Similar matrices represent the same linear transformation in different basis.

When $\mathbf{Q}$ is an orthogonal matrix, we have $\mathbf{B} = \mathbf{Q}^{\top}\mathbf{A}\mathbf{Q}$.

Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors are the “natural” basis for a linear transformation

Eigenvalues and Eigenvectors

Any linear transformation represented by $\mathbf{A} \in \mathbb{C}^{n \times n}$ has vectors that satisfy the following property,

$$\mathbf{A}\mathbf{x} = \lambda \mathbf{x}, \quad \mathbf{x} \in \mathbb{C}^n, \ \lambda \in \mathbb{C}, \ \ \mathbf{x} \neq \mathbf{0}$$

where, λ and $\mathbf{x}$ are called the eigenvalue and the associated eigenvector of $\mathbf{A}$.

Any such pair $(\lambda, \mathbf{x})$ is called the eigenpair of $\mathbf{A}$.

These are important for understanding and solving linear differential and difference equations:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) \quad \text{and} \quad \mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n]$$

Eigenvalues and Eigenvectors

Consider the differential equation, $\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) = \begin{bmatrix} -5 & 11 \\ 0 & -6 \end{bmatrix} \mathbf{x}(t)$. Let us assume that the solution is of the form, $\mathbf{x} = e^{\lambda t}\hat{\mathbf{x}}$. Then we have,

$$\frac{d\mathbf{x}(t)}{dt} = e^{\lambda t}\mathbf{A}\hat{\mathbf{x}} = e^{\lambda t}\lambda\hat{\mathbf{x}} \implies \mathbf{A}\hat{\mathbf{x}} = \lambda\hat{\mathbf{x}}$$

$$\begin{bmatrix} -5 & 11 \\ 0 & -6 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} \lambda\hat{x}_1 \\ \lambda\hat{x}_2 \end{bmatrix} \implies \begin{bmatrix} -5-\lambda & 11 \\ 0 & -6-\lambda \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

where, $\hat{\mathbf{x}} \in N(\mathbf{A} - \lambda\mathbf{I})$.

Eigenvalues and Eigenvectors

We can find the eigenpairs using the same approach for $\mathbf{A} \in \mathbb{C}^{n \times n}$,
$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 = p(\lambda)$.

$p(\lambda)$ is the characteristic polynomial of $\mathbf{A}$, and $p(\lambda) = 0$ is the characteristic equation.

The eigenvalues are the roots of the polynomial $p(\lambda)$, and the $\mathbf{x}$ in $(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = 0$ for the different λs are the corresponding eigenvectors.

Eigenvalues and Eigenvectors

Compute the eigenpairs for the following matrices: $\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Compute the eigenpairs for the following matrices: $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$.

Diagonalization of a Matrix

Diagonalization of a matrix is a change of basis to the eigenbasis

Diagonalization of a Matrix

Consider a matrix $\mathbf{A}$ with n eigenpairs $\{(\lambda_i, \mathbf{x}_i)\}_{i=1}^{n}$.

$$\mathbf{A}\begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\lambda_1\mathbf{x}_1 & \lambda_2\mathbf{x}_2 & \cdots & \lambda_n\mathbf{x}_n\end{bmatrix}$$

$$\mathbf{AX} = \mathbf{X}\begin{bmatrix}\lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n\end{bmatrix} = \mathbf{X}\tilde{}$$

If the eignevectors are linearly independent, then we have $\mathbf{X}^{-1}\mathbf{AX} = \Lambda$

Diagonalization of a Matrix

Let $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ represented by $\mathbf{A} = \begin{bmatrix} 8 & 1 \\ 2 & 7 \end{bmatrix}$. Diagonalize this matrix. What does $\mathbf{A}$ do to $\mathbf{x} = \begin{bmatrix} 3 & 4 \end{bmatrix}^\top$?

What about $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$?

Diagonalization of a Matrix

A square matrices with a complete set of eigenvectors, i.e. a linearly independent set of n eigenvectors, can be decomposed into the following,

$$\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$$

Diagonalization of a matrix: Eigenpairs of special matrices

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$$\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$$

Diagonalization of a matrix: Eigenpairs of special matrices

When $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, i.e. $\mathbf{A} = \mathbf{A}^\top$,

- All eigenvalues are real.
- The matrix poses a complete set of eigenvectors, i.e. they form a linearly independent set.
- The eigenvectors can be chosen to be orthogonal to each other. When the eigenvalues are distinct, the eigenvectors are orthogonal. But when the eigenvalues are not distinct, we can choose them to be orthogonal.

This gives us, $\mathbf{A} = \mathbf{A}^\top = \mathbf{X}\Lambda\mathbf{X}^\top$.

Diagonalization of a matrix: Eigenpairs of special matrices

Diagonalize $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Diagonalization of a matrix

A change of basis to $\mathbf{X}$ simplifies $\mathbf{A}$ to a diagonal matrix, the simplest possible form.

If a matrix $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ can always be diagonalized.

When there are repeated eigenvalues, we might not always be able to diagonalize a matrix. This happens when there aren't enough eigenvectors. These are called *defective* matrices.

$$\text{Algebraic multiplicity} \neq \text{Geometric multiplicity}$$

where, *algebraic multiplicity* is the number of times the eigenvalue λ is repeated, and *geometric multiplicity* is $\dim N\left(\mathbf{A} - \lambda \mathbf{I}\right)$.

Diagonalization of a matrix

Diagonalize $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

Jordan Form

If $\mathbf{A}$ cannot be diagonalized, the next best thing is the *Jordan form*. Let $\mathbf{A}$ have eigenvalues $(\lambda_1, \lambda_2, \ldots \lambda_k)$. We can find a similarity transformation, such that,

$$\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{J}(\lambda_1) & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \mathbf{J}(\lambda_2) & \ldots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{J}(\lambda_k) \end{bmatrix}$$

Jordan Form

Each $\mathbf{J}(\bullet)$ is associated with an eigenvalue and an eigenvector, and is called a Jordan block, and has the form

$$\mathbf{J}(\lambda_l) = \begin{bmatrix} \lambda_l & 1 & 0 & \dots & 0 \\ 0 & \lambda_l & 1 & \dots & 0 \\ 0 & 0 & \lambda_l & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \dots & \lambda_l \end{bmatrix}$$

- $\mathbf{J} \in \mathbb{C}^{r \times r}$. r = the algebraic multiplicity of the eigenvalue λ_l.
- The geometric multiplicity of the eigenvalue $\lambda_l = \dim N(\mathbf{A} - \lambda_l \mathbf{I})$.
- A 1-by-1 Jordan block is simply $[\lambda_l]$, corresponding to a eigenvalue with an associated eigenvector.

Jordan Form

- Jordan form of a diagonalizable matrix $\mathbf{A} \longrightarrow \mathbf{J} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$

$\lambda = -2$ (AM[1] = 1, GM[2] = 1), & $\lambda = 11$ (AM = 2, GM = 1) $\longrightarrow \mathbf{J} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 11 & 1 \\ 0 & 0 & 11 \end{bmatrix}$

[1] AM: Algebraic multiplicity
[2] GM: Geometric multiplicity

Jordan Form

Write down the Jordan form.

$\lambda_1 = 1$ (AM = 2, GM = 1)

$\lambda_2 = 11$ (AM = 3, GM = 2)

$\lambda_3 = 0$ (AM = 3, GM = 1)

$\lambda_4 = -1$ (AM = 2, GM = 2).